# Priors and Latent Variables

## DD2421

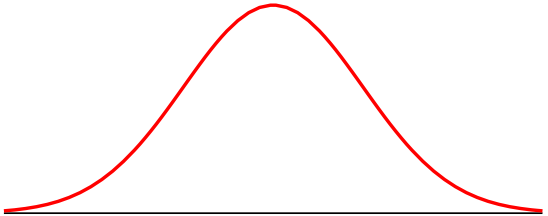
André Pereira
(slides heavily adapted from Bob Sturm)

# Summary

1. Incorporating Priors
   - Maximum a Posteriori Estimation
   - Bayesian Non-Parametric Methods
   - Model Selection and Occam's Razor
2. Unsupervised Learning
   - Classification vs Clustering
   - Heuristic Example: K-means
   - Expectation Maximization

# Problem: few data points

10 repetitions with 5 points each

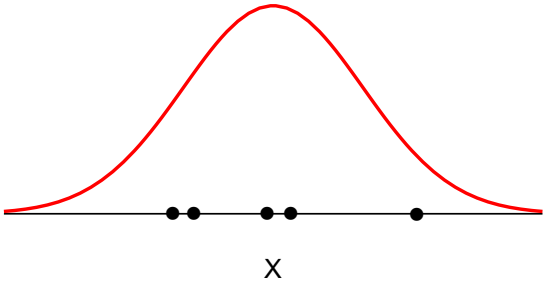


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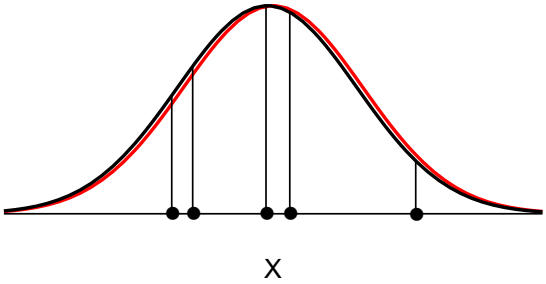
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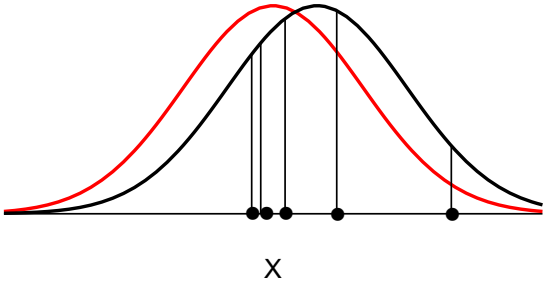
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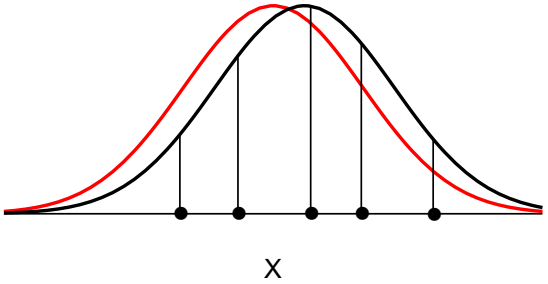
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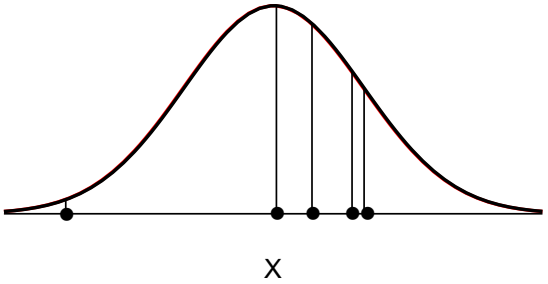
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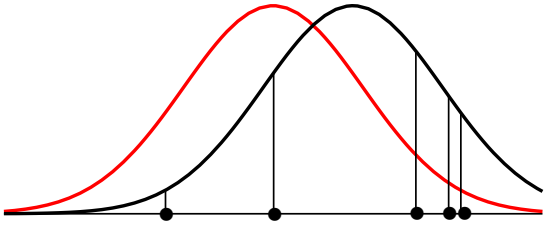
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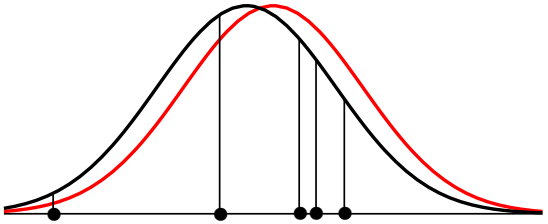
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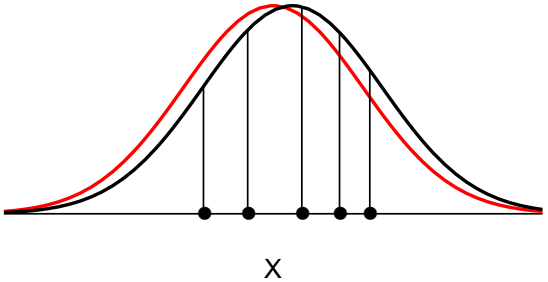
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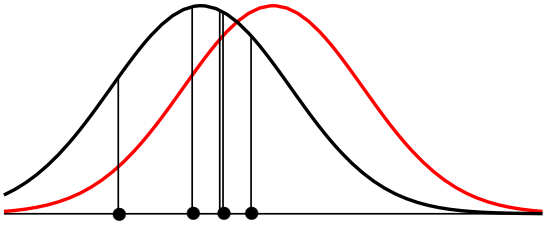
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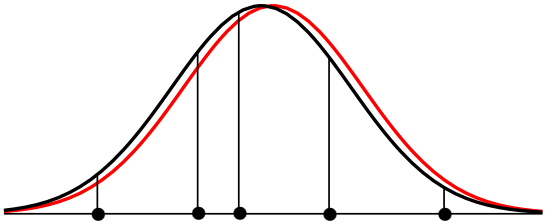
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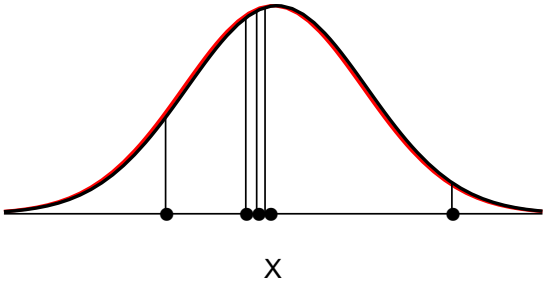
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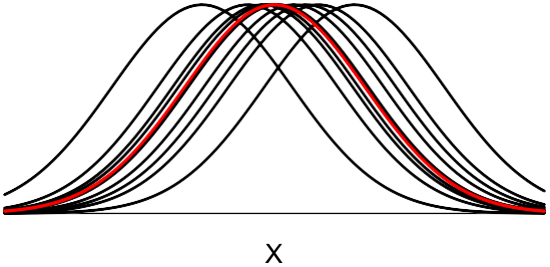
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# Maximum a Posteriori Estimation

- Recall: ML estimation chooses $\theta$ to maximize probability of $\mathcal{D}$.
- MAP estimation chooses the most likely $\theta$ given $\mathcal{D}$.

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- $\log \mathrm{p}(\theta)$ works as a regularizer.

# MAP for Linear Regression

Model (deterministic):

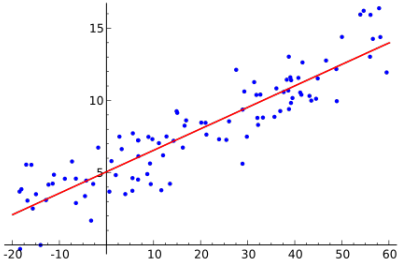
$$y = \boldsymbol{w}^T\mathbf{x} + \boldsymbol{\epsilon}$$

With:

$$\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma_Y^2)$$

Therefore:

$$\mathrm{Y}|\mathrm{X} \sim \mathcal{N}(\boldsymbol{w}^T\mathbf{x}, \sigma_Y^2)$$




But now we define the a priori probability of $\boldsymbol{w}$: $\mathrm{p}(\boldsymbol{w})$

# Example: D-dimension zero-mean spherical Gaussian prior

We assume in this case $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma_W^2 \boldsymbol{I}_D)$

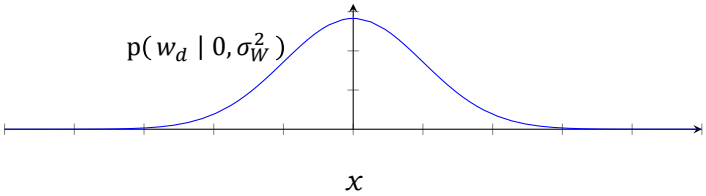
$$\mathrm{p}(\mathbf{w} \mid 0, \sigma_W^2) = \frac{1}{(2\pi\sigma_W^2)^{D/2}} \exp\left(-\frac{\mathbf{w}^T\mathbf{w}}{2\sigma_W^2}\right)$$

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$$= \prod_{d=1}^{D} \frac{1}{\sqrt{2\pi\sigma_W^2}} \exp\left(-\frac{w_d^2}{2\sigma_W^2}\right)$$



# Example: D-dimension zero-mean spherical Gaussian prior

Instead of $\log \mathrm{p}(y|x, \boldsymbol{w})$ as in MLE, we optimize $\log \mathrm{p}(\boldsymbol{w}|y, x)$:

$$\boldsymbol{w}_{\mathbf{MAP}} = \arg\max_{\mathbf{w}} \log \mathrm{p}(\mathbf{w} \mid y, x) = \arg\max_{\mathbf{w}} \log[\mathrm{p}(y \mid x, \mathbf{w})\, \mathrm{p}(\mathbf{w})]$$

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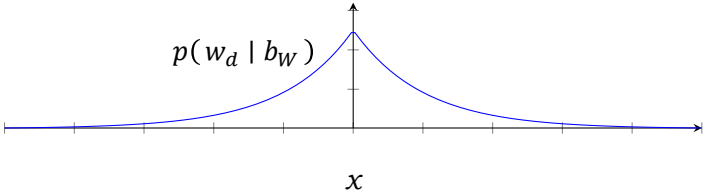
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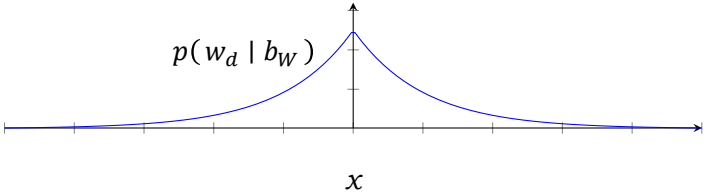
# Example: each dimension is iid zero-mean Laplace

In this case $p(\mathbf{w} \mid b_W) = \prod_d \frac{1}{2b_W} \exp\left(-\frac{|w_d|}{b_W}\right)$



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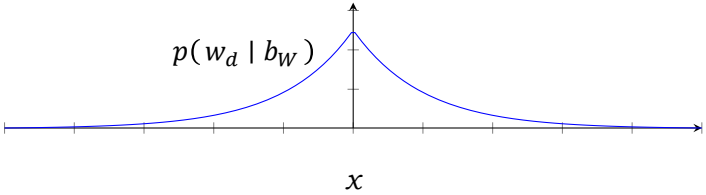
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# Impact of different priors

For linear regression $y = \mathbf{w}^{\mathbf{T}}\mathbf{x} + \epsilon$ , MAP estimation of $\mathbf{w}$ performs differently considering the prior we assume!

$$\boldsymbol{w}^{*}(\mathrm{p}) = \arg\min_{\mathbf{w}} \left[ \underbrace{\sum_{n=1}^{N} \left(y_n - \mathbf{w}^{\mathbf{T}}\mathbf{x}_{\mathbf{n}}\right)^2}_{\text{fit the data (ML)}} + \lambda \underbrace{\sum_{d=1}^{D} |w_d|^p}_{\text{regularize}} \right]$$

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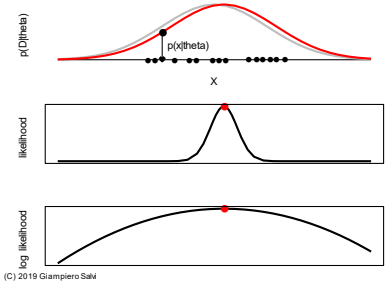
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- $\lambda > 0$ and assuming iid Normal ($p = 2$): fit the data and make **w** short (also known as *ridge regression*)
- $\lambda > 0$ and assuming iid Laplace ($p = 1$): fit the data and make **w** sparse (also know as the *Least Absolute Shrinkage and Selection Operator (LASSO)*)

# ML, MAP and Point Estimates

- Both ML and MAP produce point estimates of $\theta$
- Assumption: there is a true value for $\theta$
- advantage: once $\hat{\theta}$ found, assume everything is known



# Limitations (Linear Regression)

- With MAP estimation, the problem shifts to defining the parameters of the prior $\mathrm{p}(\boldsymbol{w})$ (equivalently choosing $\lambda$ in ridge or LASSO regression)
- We still have uncertainty in the posterior $\mathrm{p}(y|\boldsymbol{x}, \boldsymbol{w}^*)$ but this is not explicit.
- We don't want $\mathrm{p}(y|\boldsymbol{x}, \boldsymbol{w})$, but instead $\mathrm{p}(y|\boldsymbol{x}, D)$.

# Bayesian estimation (non-parametric models)

| | | | | | |
|---|---|---|---|---|---|
| ML: | $\mathcal{D}$ | $\rightarrow$ | $\theta_{\mathrm{ML}}$ | $\rightarrow$ | $\mathrm{p}(y\|\mathrm{x},\theta_{\mathrm{ML}})$ |
| MAP: | $\mathcal{D}, \mathrm{p}(\theta)$ | $\rightarrow$ | $\theta_{\mathrm{MAP}}$ | $\rightarrow$ | $\mathrm{p}(y\|\mathrm{x},\theta_{\mathrm{MAP}})$ |
| Bayes: | $\mathcal{D}, \mathrm{p}(\theta)$ | $\rightarrow$ | $\mathrm{p}(\theta\|\mathcal{D})$ | $\rightarrow$ | $\mathrm{p}(y\|\mathrm{x},\mathcal{D})$ |

1. consider $\theta$ as a random variable (same as MAP)
2. characterize $\theta$ with the posterior distribution $\mathrm{p}(\theta|\mathcal{D})$ given the data
3. compute new predicting posterior $\mathrm{p}(y|x,\mathcal{D})$ marginalizing over $\theta$ (predictive posterior)

$$p(\,y \mid \mathrm{x},\mathcal{D}\,) = \int_{\theta\in\Theta} p(\,y \mid \mathrm{x},\theta\,)\; p(\,\theta \mid \mathcal{D}\,)\, d\theta$$

# Bayesian Linear Regression

Setup:

$$\mathcal{D} = \{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_N, y_N)\}$$

Model (same as MAP):

- $Y_1, \ldots, Y_n$ conditionally independent given $\mathbf{w}$
- $\mathrm{Y}|\mathrm{X} \sim \mathcal{N}(\boldsymbol{w}^T \mathbf{x}, \sigma_Y^2)$
- $\boldsymbol{w} \sim \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{W}}, \boldsymbol{\Sigma}_{\boldsymbol{W}}), \quad \boldsymbol{w} \in \mathbb{R}^{\mathcal{D}}$
- we assume values for $\sigma_Y^2$, $\boldsymbol{\mu}_W$, and $\boldsymbol{\Sigma}_{\boldsymbol{W}}$ , so $\theta \equiv \boldsymbol{w}$

Goal: Estimate $\mathrm{p}(y|\boldsymbol{x}, \mathcal{D})$

# Bayesian Linear Regression

$$\mathrm{p}(y \mid \mathbf{x}, \mathcal{D}) = \int_{\mathbb{R}^{\mathcal{D}}} \mathrm{p}(y \mid \mathbf{x}, \mathcal{D}, \mathbf{w}), \mathrm{p}(\mathbf{w} \mid \mathbf{x}, \mathcal{D}), d\mathbf{w}$$

$$= \int_{\mathbb{R}^{\mathcal{D}}} \mathrm{p}(y \mid \mathbf{x}, \mathbf{w}) \, \mathrm{p}(\mathbf{w} \mid \mathcal{D}) d\mathbf{w}$$

(assuming $Y|\boldsymbol{W} = \boldsymbol{w}$ is independent of $\mathcal{D}$, and $\boldsymbol{W}$ is independent of $\mathbf{x}$). Results obtained with a lot of passages:

- if $\mathrm{p}(\boldsymbol{w})$ is Gaussian, then $\mathrm{p}(\boldsymbol{w}|D)$ is Gaussian
- since $\mathrm{p}(y|\boldsymbol{x}, \boldsymbol{w})$ is Gaussian, $\mathrm{p}(y|\boldsymbol{x}, D)$ is also Gaussian
- Closed-form results in this case.

# Find a walk-through of the derivations here

From mathematicalmonk's YouTube channel:

- problem and model definition
  https://youtu.be/1Wvnpjljk XA
- posterior $p(\boldsymbol{w}|D)$, part 1–2
  https://youtu.be/nrd4AnDLR3U
  https://youtu.be/qz2U8coNwV4
- predictive posterior $p(y|\boldsymbol{x}, D)$, part 1–3
  https://youtu.be/xyuSiKXttxw
  https://youtu.be/vTcsacTqlfQ
  https://youtu.be/LCISTY9S6SQ

# Closed Form Solutions

Define $\Phi^T = [\,\mathbf{x_1}\ \mathbf{x_2}\ \cdots\ \mathbf{x_N}\,]$ and $\boldsymbol{y} = (y_1, y_2, \ldots, y_N)$ from $\mathcal{D}$.
With the assumed set-up, the posterior $\mathrm{W}|\mathcal{D} \sim \mathcal{N}(\boldsymbol{\mu_{W|\mathcal{D}}}, )$, with:

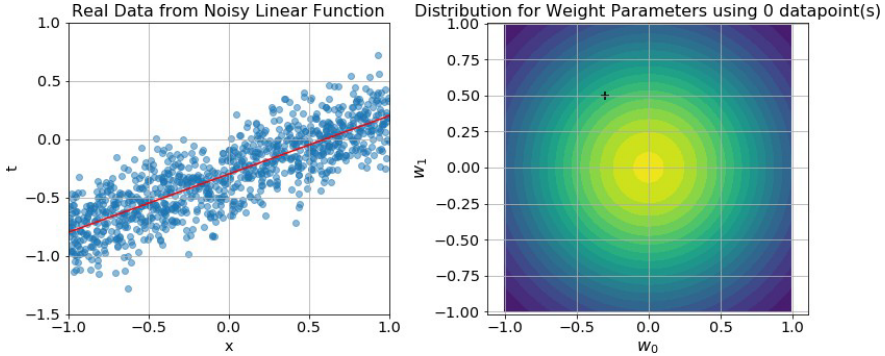
$$\Sigma_{\mathbf{w}|\mathcal{D}} = \left[\Sigma_{\mathbf{w}}^{-1} + \frac{1}{\sigma_Y^2}\Phi^{\mathbf{T}}\Phi\right]^{-1}$$

$$\mu_{\mathbf{w}|\mathcal{D}} = \Sigma_{\mathbf{w}|\mathcal{D}}\left[\Sigma_{\mathbf{w}}^{-1}\mu_{\mathbf{w}} + \frac{1}{\sigma_Y^2}\Phi^{\mathbf{T}}\mathbf{y}\right]$$

Predictive posterior:

$$P(y|\mathbf{x},\mathcal{D}) = \mathcal{N}\,(\mu_{\mathbf{w}|\mathcal{D}}^T\mathbf{x}, \sigma_Y^2 + \mathrm{x}^T\Sigma_{\mathbf{w}|\mathcal{D}}\mathbf{x})$$

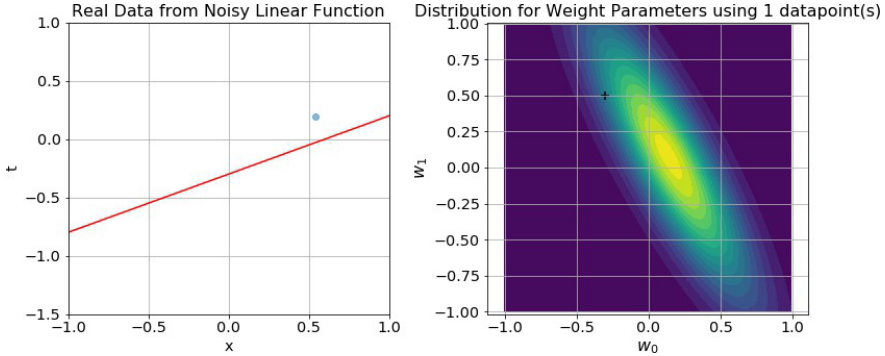
# Bayesian Linear Regression: Example



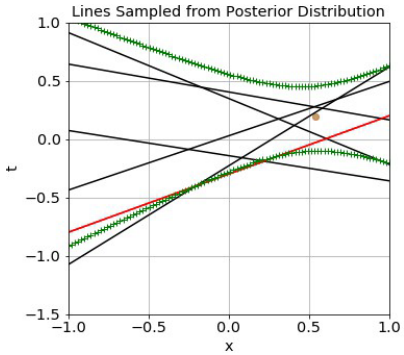

Largely adapted from

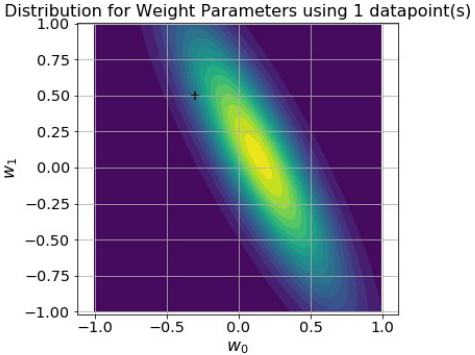
https://zjost.github.io/bayesian-linear-regression/

# Bayesian Linear Regression: Example

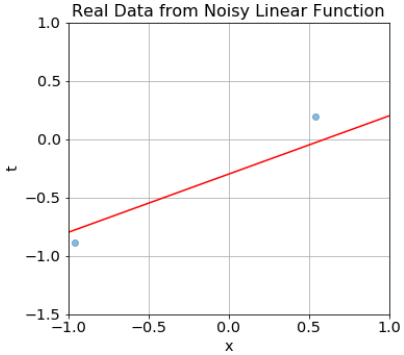


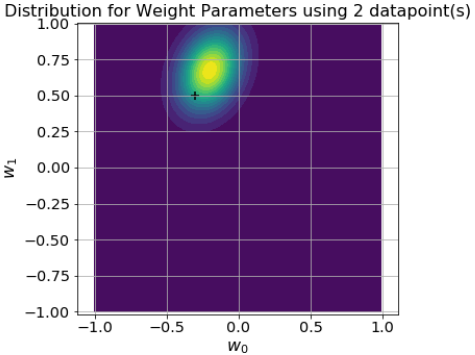
# Bayesian Linear Regression: Example



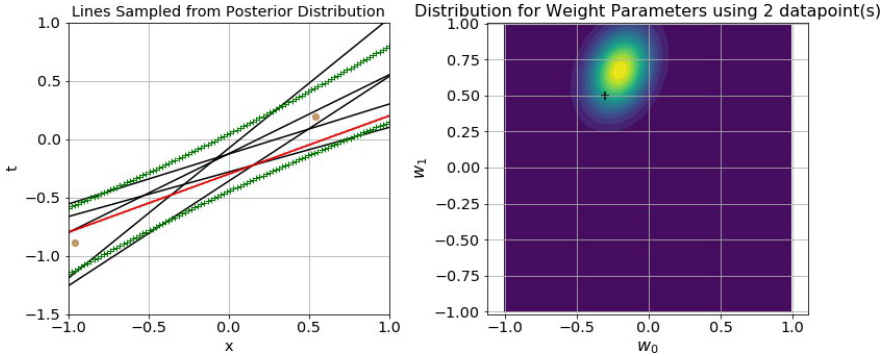



# Bayesian Linear Regression: Example

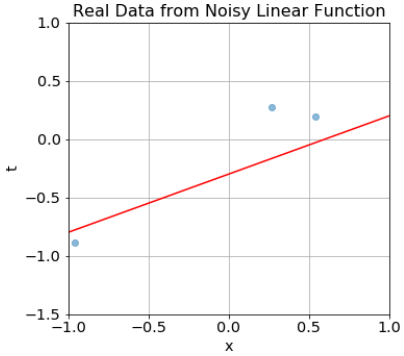


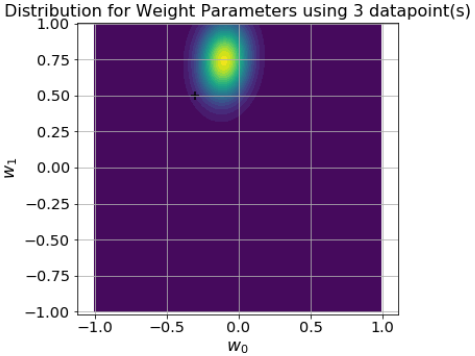



# Bayesian Linear Regression: Example

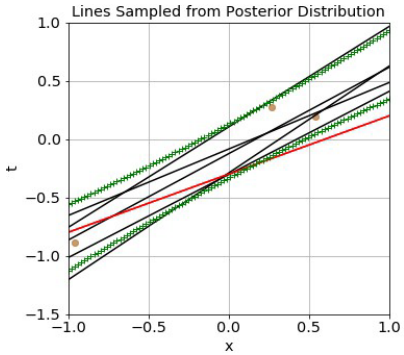


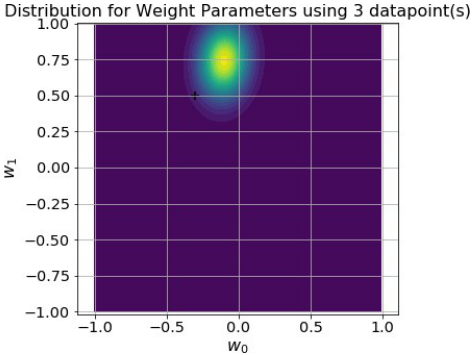
# Bayesian Linear Regression: Example



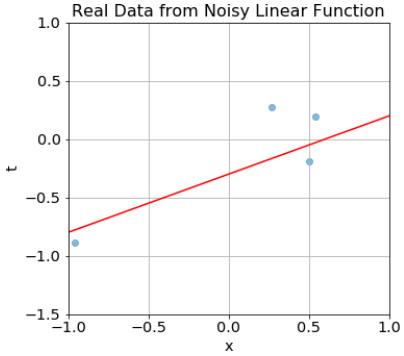



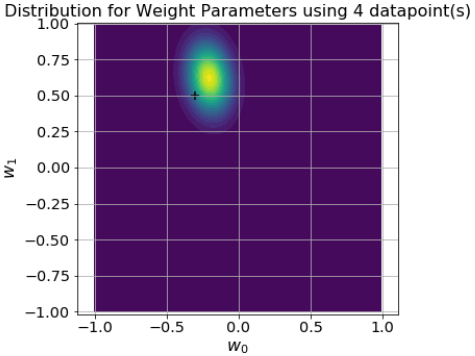
# Bayesian Linear Regression: Example



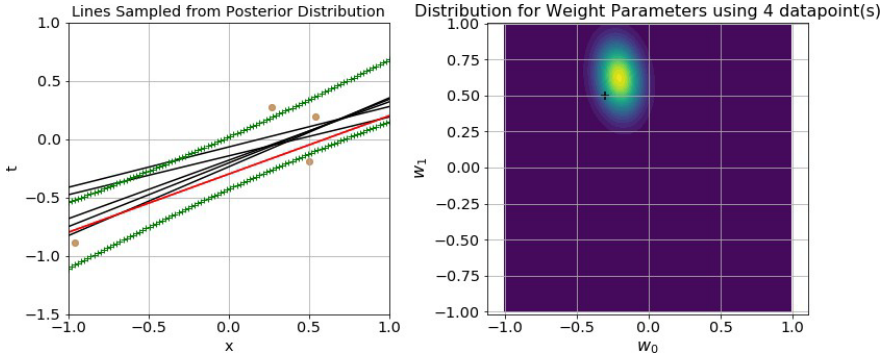



# Bayesian Linear Regression: Example

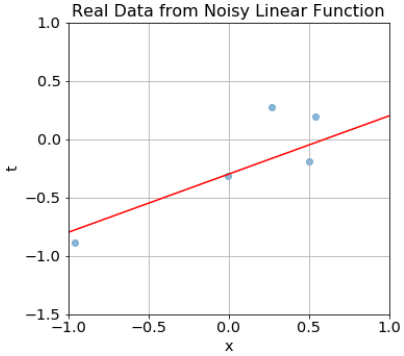


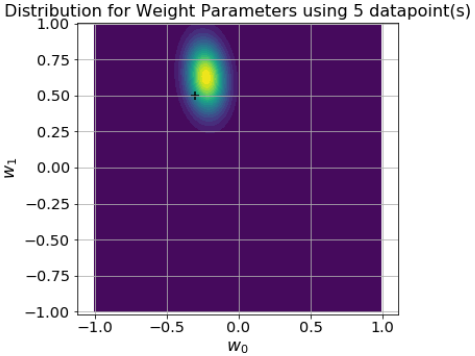



# Bayesian Linear Regression: Example

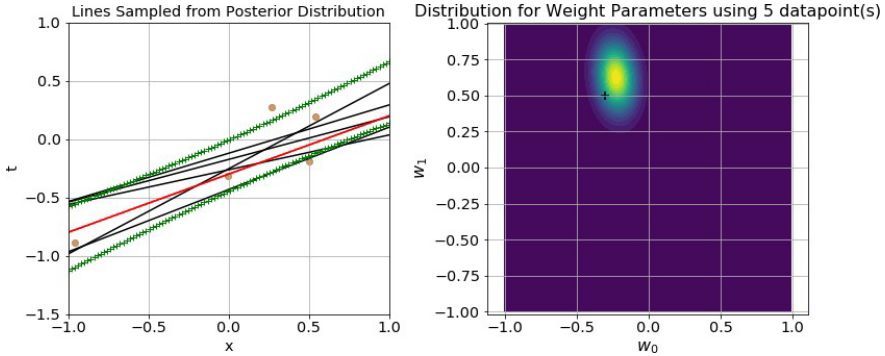


# Bayesian Linear Regression: Example

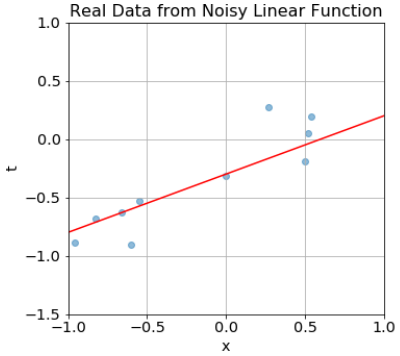


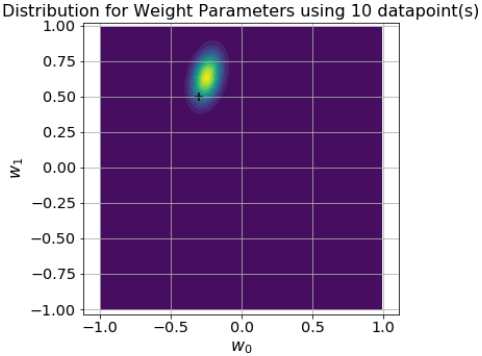



# Bayesian Linear Regression: Example

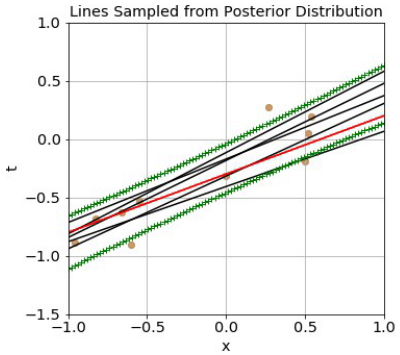


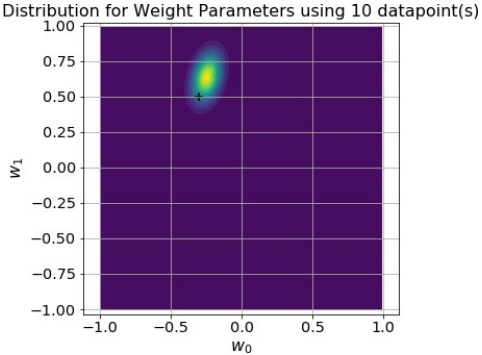
# Bayesian Linear Regression: Example



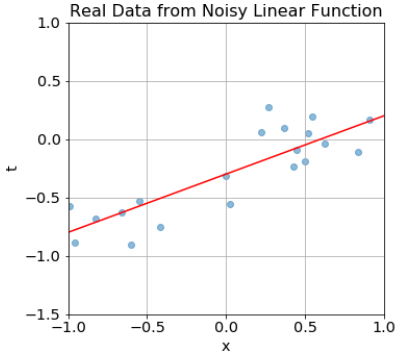



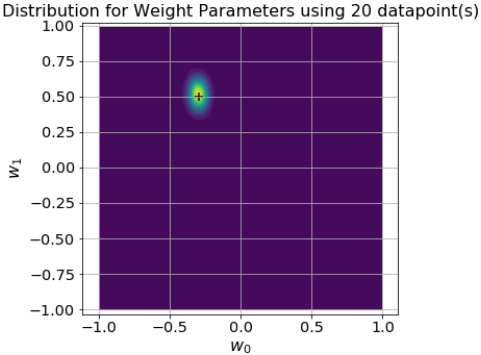
# Bayesian Linear Regression: Example



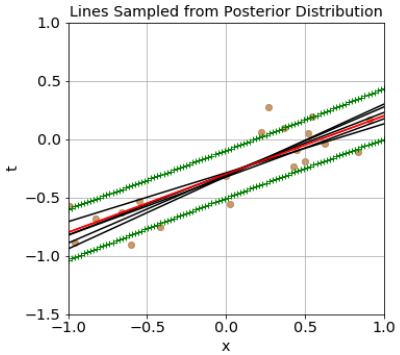



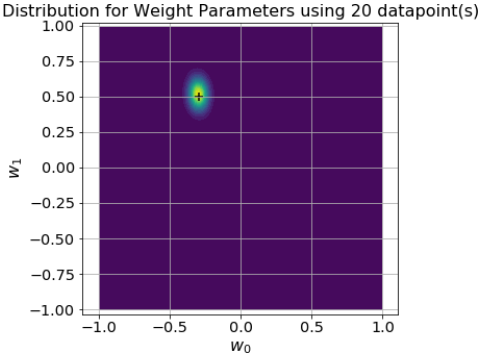
# Bayesian Linear Regression: Example



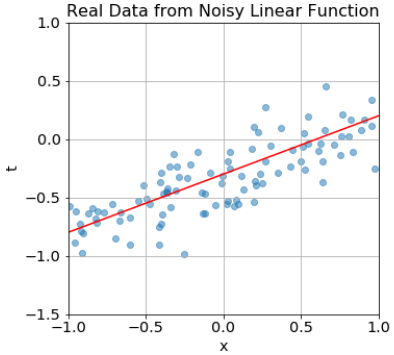



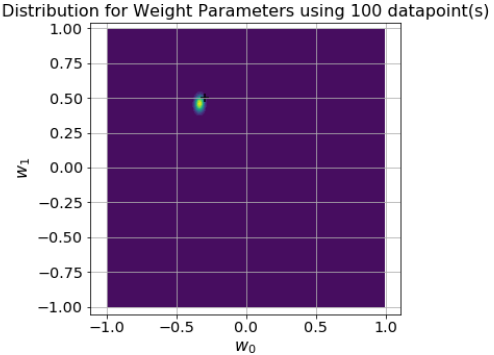
# Bayesian Linear Regression: Example



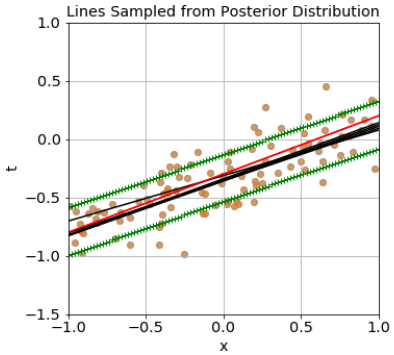



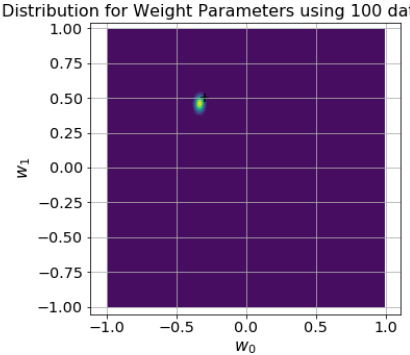
# Bayesian Linear Regression: Example



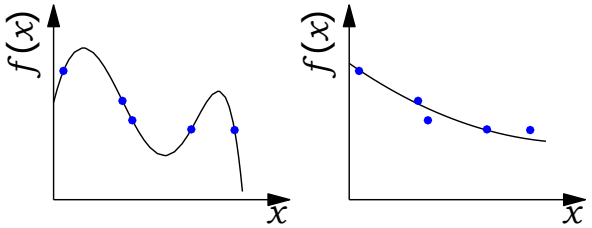
# Bayesian Linear Regression: Example



# Consequences

- We are considering the uncertainty over the choice of **w** as well as the original uncertainty $\sigma_Y^2$.
- This provides a means to prevent overfitting



# Occam's Razor

*Choose the simplest explanation for the observed data*

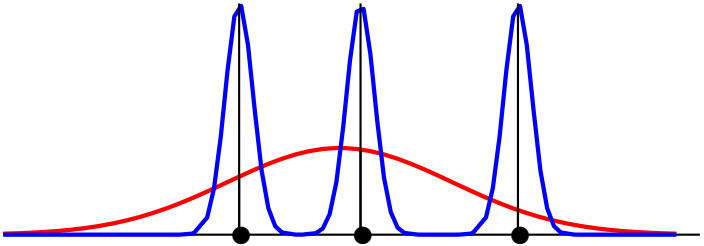
Important factors:

- number of model parameters
- number of data points
- model fit to the data

# Overfitting and Maximum Likelihood

$$\theta_{ML} = \arg\max_{\theta} P(\mathcal{D}|\theta)$$

We can make the likelihood arbitrary large by increasing the number of parameters



# Occam's Razor and Bayesian Learning

Remember that:

$$p(y \mid \mathbf{x}, \mathcal{D}) = \int_{\theta \in \Theta} p(y \mid \mathbf{x}, \theta)\, p(\theta \mid \mathcal{D})\, d\theta$$

# Occam's Razor and Bayesian Learning

Remember that:

$$p(y \mid \mathbf{x}, \mathcal{D}) = \int_{\theta \in \Theta} p(y \mid \mathbf{x}, \theta)\, p(\theta \mid \mathcal{D})\, d\theta$$

**Intuition:**

More complex models fit the data very well (large $p(\mathcal{D}|\theta)$ and $p(\theta|\mathcal{D})$) but only for small regions of the parameter space $\Theta$.

# Limitations of Bayesian Non-parametric Methods

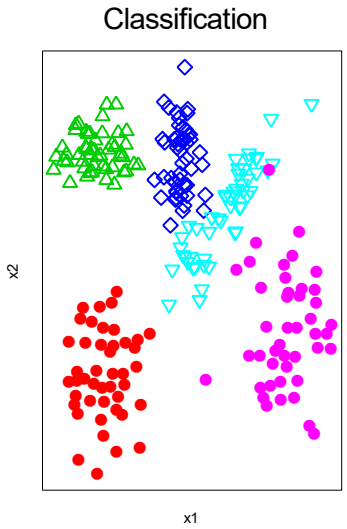
$$p(y \mid \mathbf{x}, \mathcal{D}) = \int_{\mathbb{R}^{\mathcal{D}}} \mathrm{p}(y \mid \mathbf{x}, \mathbf{w})\, \mathrm{p}(\mathbf{w} \mid \mathcal{D})\, d\mathbf{w}$$

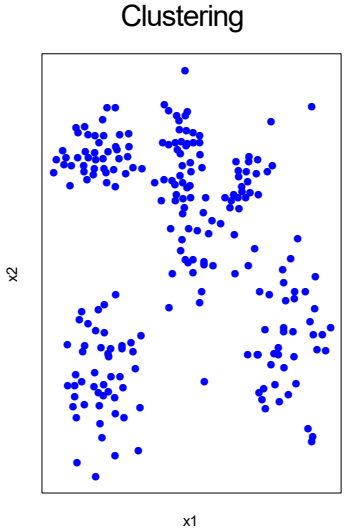
- closed form solution for $\mathrm{p}(\boldsymbol{w} \mid \mathcal{D})$ is not always possible (conjugate priors)
- can use approximations with high computational cost (sampling methods) or complex solutions (variational methods)
- sometimes we will have a non-informative prior of $\boldsymbol{W}$, but Bayesian methods carry uncertainty estimates.

# Outline

1. Incorporating Priors
   - Maximum a Posteriori Estimation
   - Bayesian Non-Parametric Methods
   - Model Selection and Occam's Razor
2. Unsupervised Learning
   - Classification vs Clustering
   - Heuristic Example: K-means
   - Expectation Maximization

# Clustering vs Classification



# Heuristic Example: K-means

- describes each class with a centroid
- a point belongs to a class if the corresponding centroid is closest (Euclidean distance)
- iterative procedure
- guaranteed to converge
- not guaranteed to find the optimal solution used
- in *vector quantization* (since the 1950's)

# K-means: algorithm

**Data:** No. of desired clusters $K$, data set $D = \{\boldsymbol{x}_i \in \mathbb{R}^{\mathcal{D}}\}$,
distance function $f : \mathbb{R}^{\mathcal{D}} \times \mathbb{R}^{\mathcal{D}} \rightarrow \mathbb{R}_+$, threshold $\epsilon$

initialization: sample $K$ centroids $\boldsymbol{\mu}_k^{(0)} \in \mathbb{R}^{\mathcal{D}}$ at random; $t \leftarrow 0$;

**repeat**

assign each point $x_i$ to closest centroid:

$$C_k \coloneqq \{\mathbf{x} \in \mathcal{D}: k = \arg\min_{k'} f\left(\mathbf{x}, \boldsymbol{\mu}_{k'}^{(t)}\right)\}$$

update centroids as mean of each cluster of points:

$$\boldsymbol{\mu}_k^{(t+1)} = \frac{1}{|C_k|} \sum C_k$$

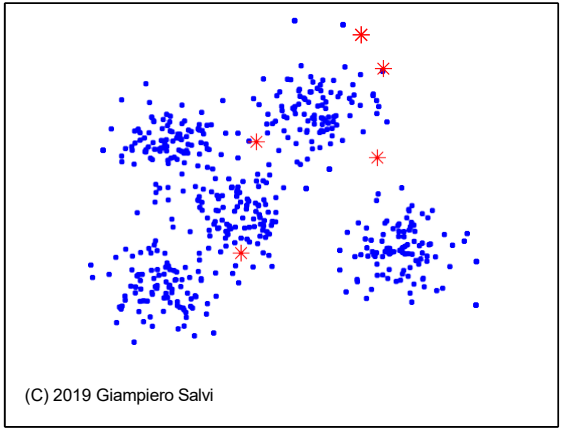
iterate $t \leftarrow t + 1$

**until** $max_k\, f\,(\boldsymbol{\mu}_k^{(t-1)}, \boldsymbol{\mu}_k^{(t)}) < \epsilon$;

**return** $K$ clusters of data points.

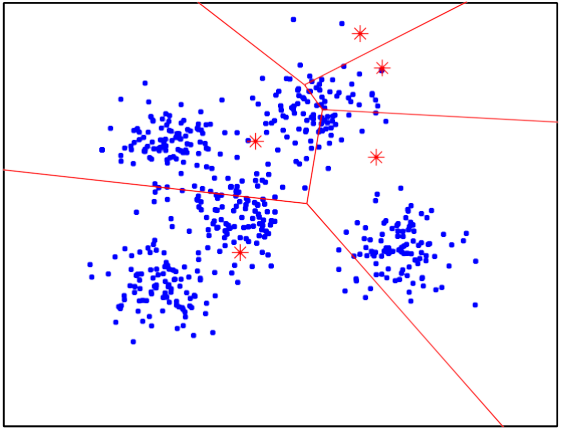
# K-means: example

initialization



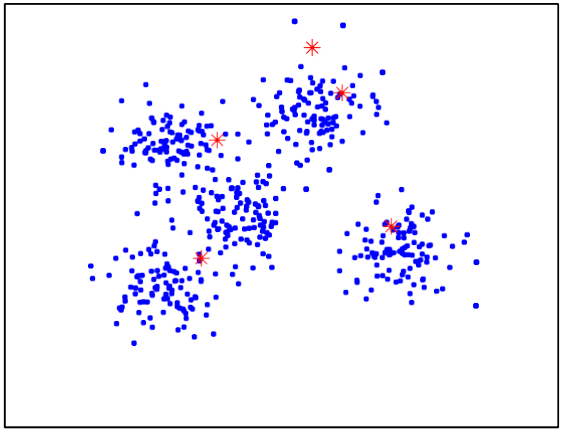
# K-means: example

iteration 1, update clusters



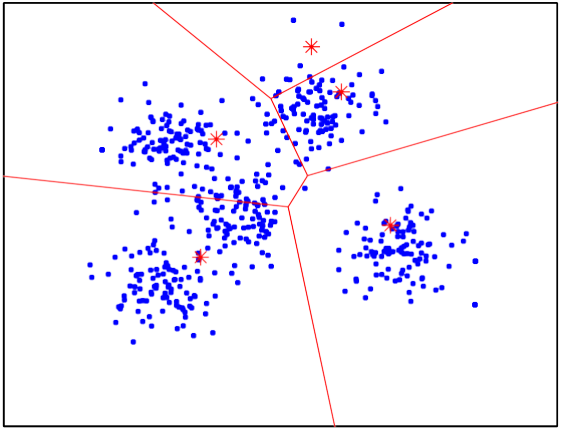
# K-means: example

iteration 2, update centroids



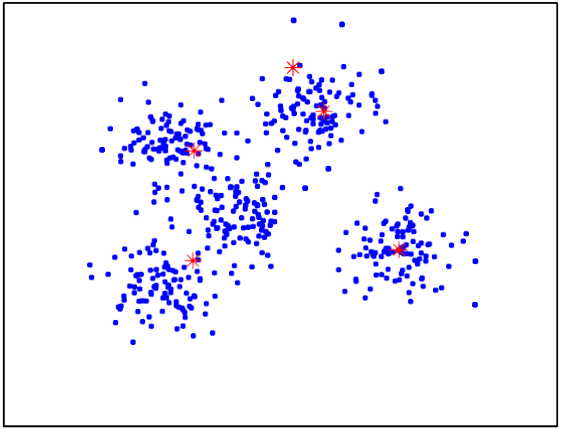
# K-means: example

iteration 2, update clusters



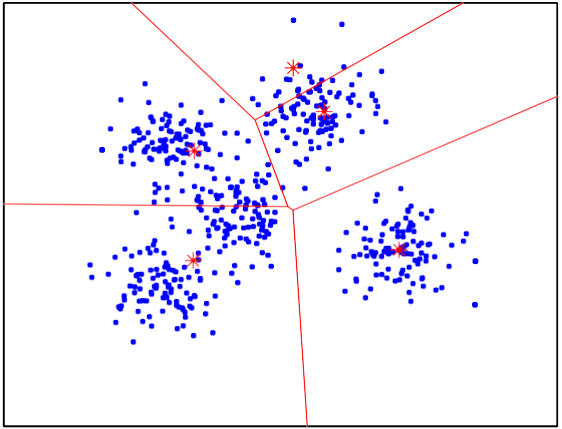
# K-means: example

iteration 3, update centroids



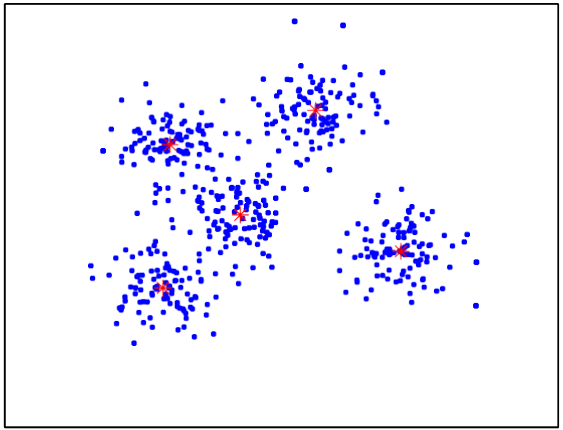
# K-means: example

iteration 3, update clusters



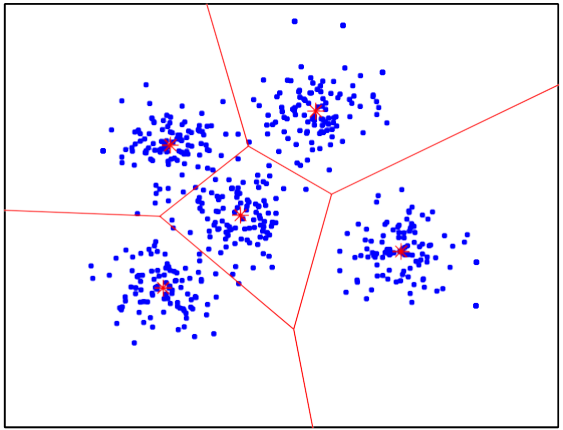
# K-means: example

iteration 20, update centroids



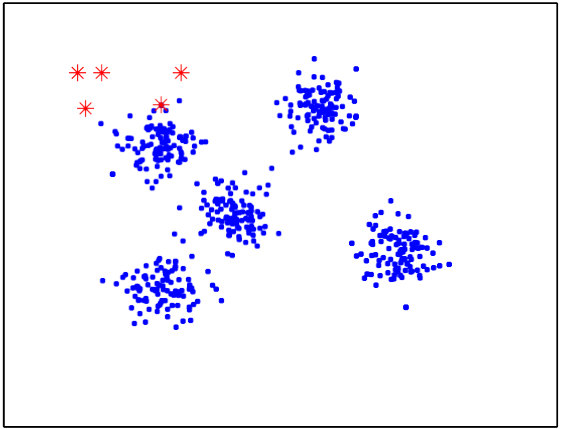
# K-means: example

iteration 20, update clusters



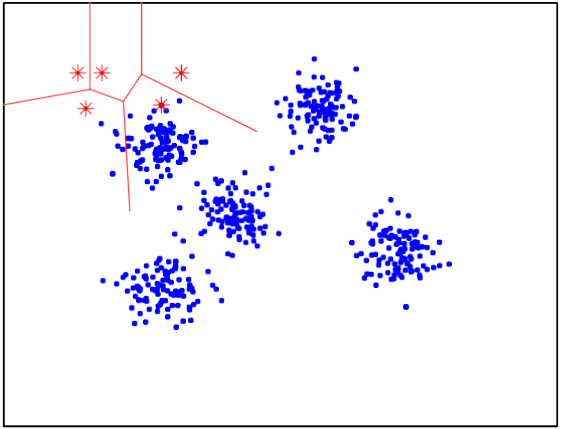
# K-means: sensitivity to initial conditions

initialization



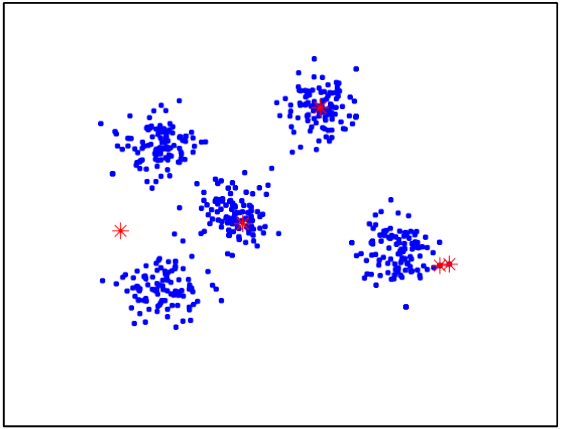
# K-means: sensitivity to initial conditions

iteration 1, update clusters



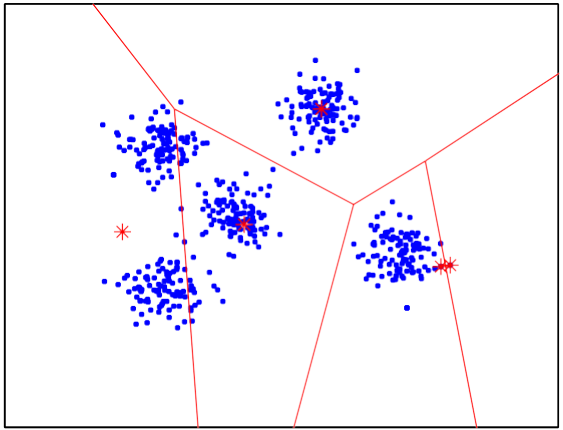
# K-means: sensitivity to initial conditions

iteration 2, update centroids



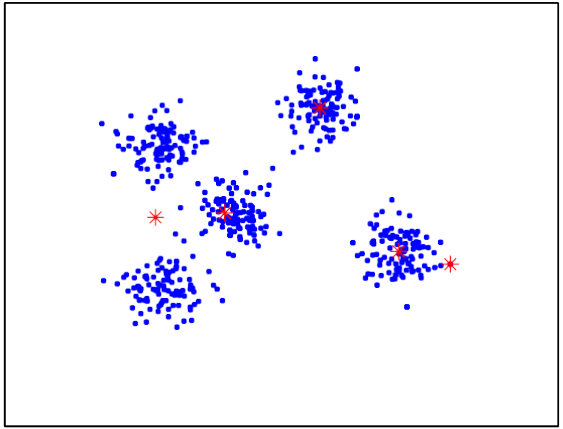
# K-means: sensitivity to initial conditions

iteration 2, update clusters



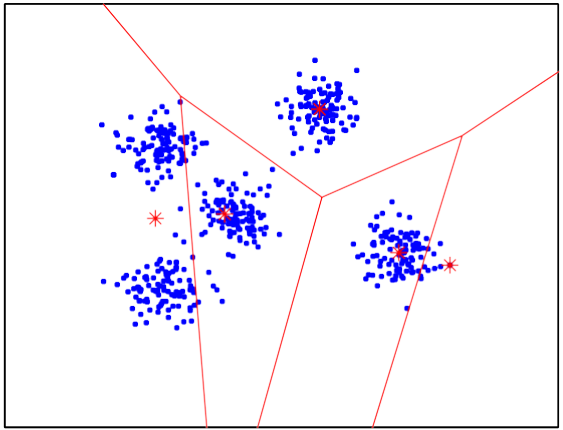
# K-means: sensitivity to initial conditions

iteration 3, update centroids



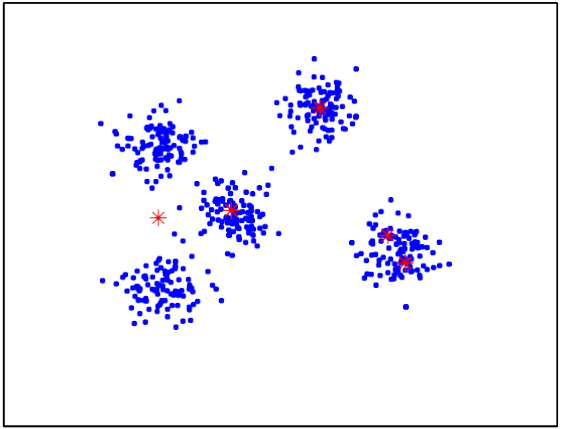
# K-means: sensitivity to initial conditions

iteration 3, update clusters



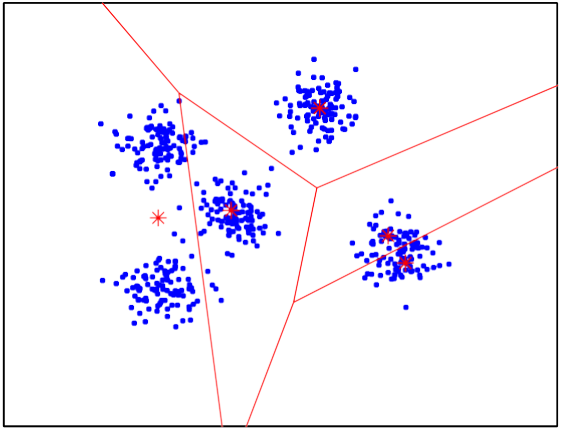
# K-means: sensitivity to initial conditions

iteration 20, update centroids



# K-means: sensitivity to initial conditions

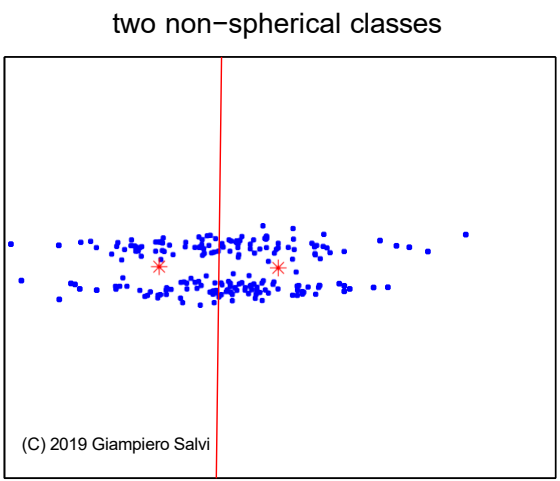
iteration 20, update clusters



# K-means: limits of Euclidean distance

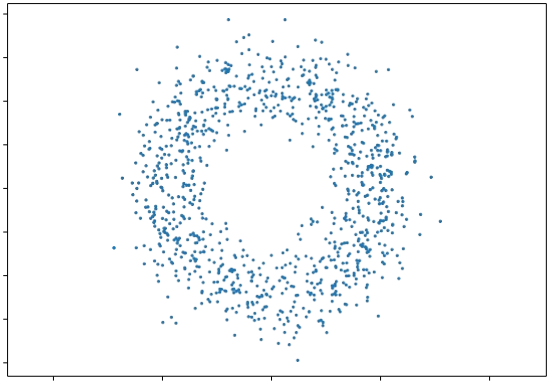
- the Euclidean distance is isotropic (same in all directions in $\mathbb{R}^p$)
- this favors spherical clusters
- the size of the clusters is controlled by the distances between them

# K-means: non-spherical classes



# Example: doughnut data

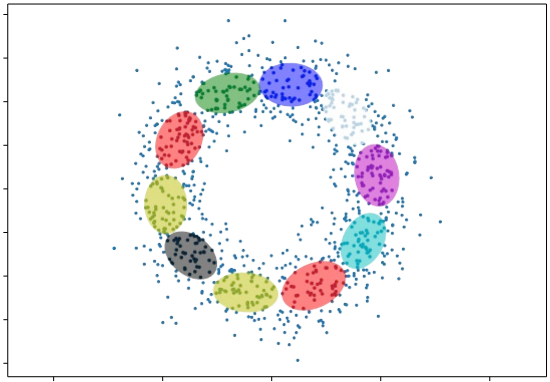
$$\mathcal{D} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$$



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# Example: doughnut data

$$\mathcal{D} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$$
$$\mathrm{p}(\mathbf{x} \mid \theta_k), \ \forall k \in [1, K]$$

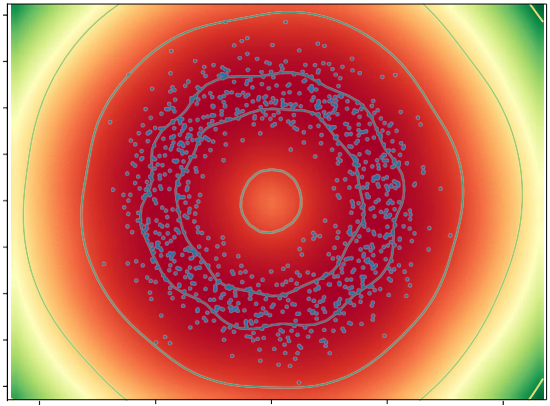


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# Example: doughnut data

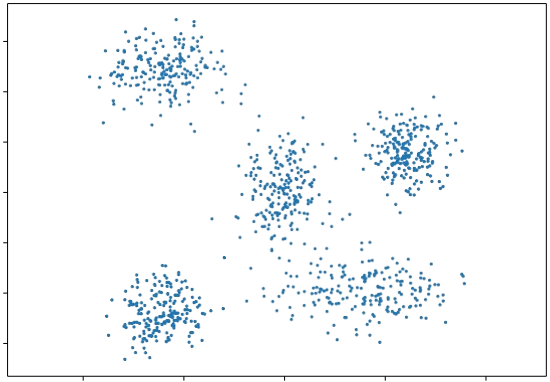
$$\mathcal{D} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$$

$$p(\mathbf{x} \mid \theta) = \sum_{k=1}^{K} \pi_k \; p(\mathbf{x} \mid \theta_k)$$



# Clustering Example

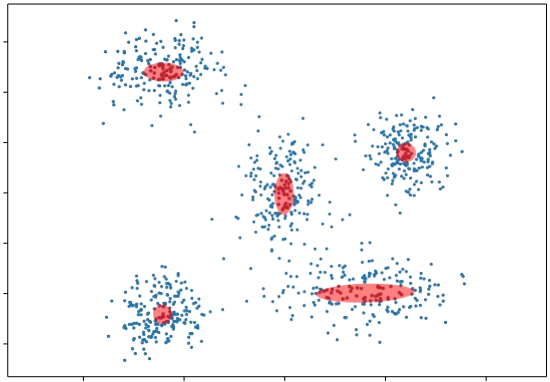
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# Clustering Example

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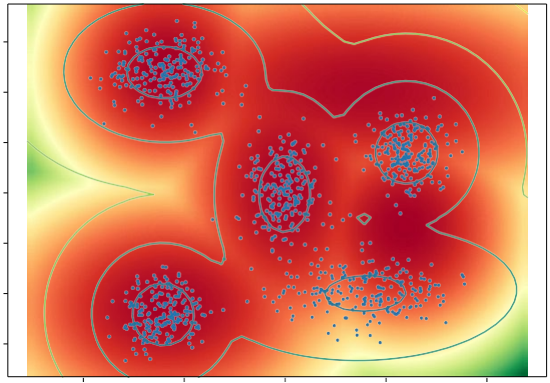


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# Clustering Example

$$\mathcal{D} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$$

$$p(\mathbf{x} \mid \theta) = \sum_{k=1}^{K} \pi_k \; p(\mathbf{x} \mid \theta_k)$$



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# Fitting complex distributions

We can try to fit a mixture of $K$ distributions:

$$p(\mathbf{x} \mid \theta) = \sum_{k=1}^{K} \pi_k \; p(\mathbf{x} \mid \theta_k),$$

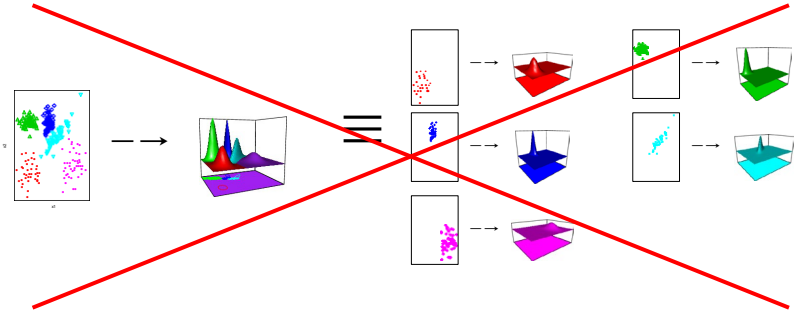
with $\theta = \{\pi_1, \ldots, \pi_k, \theta_1, \ldots, \theta_k\}$

**Problem:**

We do not know which point has been generated by which component of the mixture

We cannot optimize $p(\mathbf{x} \mid \theta)$ directly

# No Class Independence Assumption



Solution: Expectation Maximization

# Expectation Maximization

Fitting model parameters with missing (latent) variables

$$p(\mathbf{x} \mid \theta) = \sum_{k=1}^{K} \pi_k \; p(\mathbf{x} \mid \theta_k),$$

$$\text{with } \theta = \{\pi_1, \ldots, \pi_K, \theta_1, \ldots, \theta_K\}$$

- very general idea (applies to many different probabilistic models)
- augment data with latent variables: $h_i \in \{1, \ldots, K\}$ is the assignment of data point $\mathbf{x}_i$ to a component of the mixture
- optimize likelihood of the complete data over $N$ data points

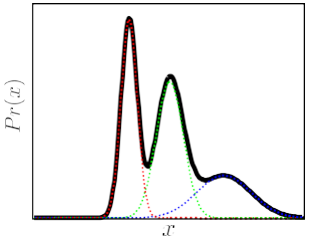
$$\mathrm{p}(\mathbf{x}_1, \ldots, \boldsymbol{x}_N, h_1, \ldots, h_N \mid \theta)$$

# Example: Mixture of Gaussians

This distribution is a weighted sum of $K$ Gaussian distributions

$$p(\mathrm{x}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}\left(\mathrm{x} \mid \mu_k, \sigma_k^2\right),$$

where $\pi_1 + \cdots + \pi_K = 1, \pi_k > 0 \, (k = 1, \ldots, K)$.



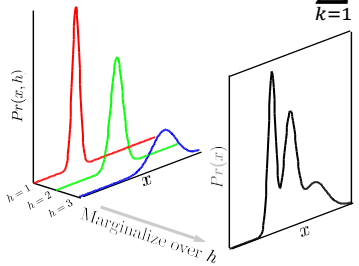

This model can describe **complex multi-modal** probability distributions by combining simpler distributions.

# Mixture of Gaussians as a marginalization

We can interpret the Mixture of Gaussians model with the introduction of a discrete hidden/latent variable $h$ and $Pr(x, h)$:

$$p(x) = \sum_{k=1}^{K} \mathrm{p}(x, h = k) = \sum_{k=1}^{K} \mathrm{p}(x, h = k)\, p(h = k)$$

$$= \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathrm{x}|\mu_k, \sigma_k^2)$$




← **mixture density**

Figures taken from **Computer Vision: models, learning and inference** by Simon Prince.

# EM for two Gaussians

For each sample $x_i$ introduce a *hidden variable* $h_i$

$$h_i = \begin{cases} 1 & \text{if sample } x_i \text{ was drawn from } \mathcal{N}(\mu_1, \sigma_1^2) \\ 2 & \text{if sample } x_i \text{ was drawn from } \mathcal{N}(\mu_2, \sigma_2^2) \end{cases}$$

and come up with initial values

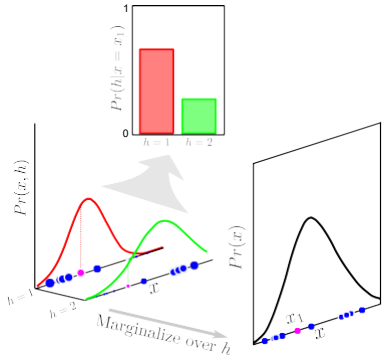
$$\Theta^{(0)} = \left( \pi_1^{(0)}, \mu_1^{(0)}, \sigma_1^{(0)}, \mu_2^{(0)}, \sigma_2^{(0)} \right)$$

for each of the parameters.

EM is an *iterative algorithm* which updates $\Theta^{(t)}$ using the following two steps...

# EM for two Gaussians: E-step

The responsibility of *k*-th Gaussian for each sample *x* (indicated by the size of the projected data point)




**Look at each sample *x* along hidden variable *h* in the E-step**

Figure from **Computer Vision: models, learning and inference** by Simon Prince.

# EM for two Gaussians: E-step (cont.)

**E-step:** Compute the posterior probability (*responsibility*) that $x_i$ was generated by component $k$ given the current estimate of the parameters $\Theta^{(t)}$ .

for each data point $i = 1, \ldots, N$

for each mixture component $k = 1, 2$

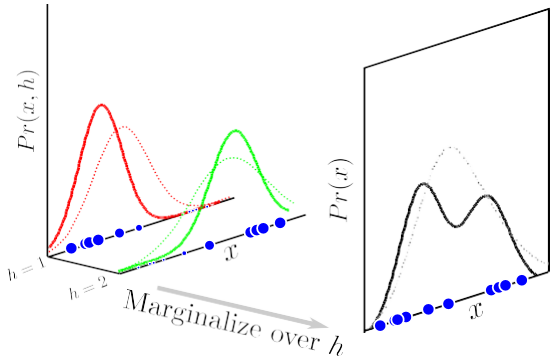
$$\gamma_{ik}^{(t)} \coloneqq \mathrm{p}\left(h_i = k \middle| x_i, \Theta^{(t)}\right) = \frac{\mathrm{p}\left(h_i = k, \Theta^{(t)}\right)\mathrm{p}\left(x_i \middle| h_i = k, \Theta^{(t)}\right)}{\mathrm{p}\left(x_i \middle| \Theta^{(t)}\right)}$$

$$= \frac{\pi_k^{(t)} \mathcal{N}\left(x_i \middle| \mu_k^{(t)}, \sigma_k^{(t)}\right)}{\pi_1^{(t)} \mathcal{N}\left(x_i \middle| \mu_1^{(t)}, \sigma_1^{(t)}\right) + \pi_2^{(t)} \mathcal{N}\left(x_i \middle| \mu_2^{(t)}, \sigma_2^{(t)}\right)}$$

**Note:** *Responsibilities* $\gamma_{i1}^{(t)} + \gamma_{i2}^{(t)} = 1$ and mixture weights $\pi_1 + \pi_2 = 1$

# EM for two Gaussians: M-step

Fitting the Gaussian model for each of *k*-th constituent.
Sample $x_i$ contributes according to the *responsibility* $\gamma_{ik}$.




(dashed and solid lines for fit before and after update)

**Look along samples *x* for each *h* in the M-step**

# EM for two Gaussians: M-step (cont.)

**M-step:** Compute the maximum likelihood estimates of $\Theta^{(t)}$ given the data membership distributions (the $\gamma_{i1}^{(t)}, \gamma_{i2}^{(t)}$):

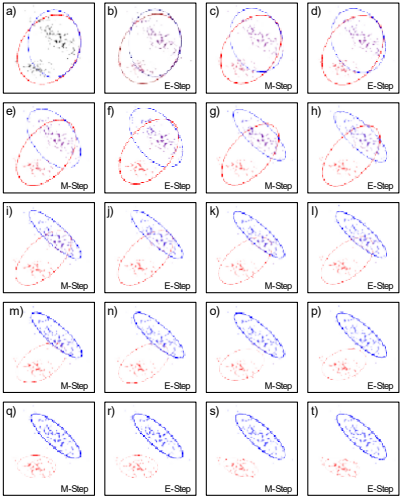
for $k = 1, 2$

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^{N} \gamma_{ik}^{(t)} x_i}{\sum_{i=1}^{N} \gamma_{ik}^{(t)}}$$

$$\sigma_k^{(t+1)} = \sqrt{\frac{\sum_{i=1}^{N} \gamma_{ik}^{(t)} \left(x_i - \mu_k^{(t+1)}\right)^2}{\sum_{i=1}^{N} \gamma_{ik}^{(t)}}}$$

$$\pi_k^{(t+1)} = \frac{\sum_{i=1}^{N} \gamma_{ik}^{(t)}}{N}$$

# EM in practice



# EM properties

Similar to K-means

- guaranteed to find a local maximum of the complete data likelihood
- somewhat sensitive to initial conditions

Better than K-means

- Gaussian distributions can model clusters with different shapes
- all data points are smoothly used to update all parameters
- Can include a prior $p(\Theta)$ to introduce regularization

# Summary

1. Incorporating Priors
   - Maximum a Posteriori Estimation
   - Bayesian Non-Parametric Methods
   - Model Selection and Occam's Razor
2. Unsupervised Learning
   - Classification vs Clustering
   - Heuristic Example: K-means
   - Expectation Maximization